

Research Interests

Birational geometry — minimal model program, Sarkisov program, foliations. AI4Math: human verification of AI-generated results, Lean 4 formalization, training approaches for math-specialized models, research-harness building.

Experience

Education

2024 – now **Ph.D. in Mathematics**, AMSS–UCAS (in progress; expected June 2027)

2022 – 2024 **M.S. in Mathematics**, AMSS–UCAS

2016 – 2020 **B.S. in Mathematics**, Beihang University

Honors

MCM/ICM Meritorious Winner (twice); Hua Luogeng Scholarship, Beihang (twice); CAS Merit Student.

Talks

Speaker, **Sarkisov Program for Algebraically Integrable and Threefold Foliations**, Xi'an Jiaotong-Liverpool University, 2026.

Mathematics

Publications

(1) Sarkisov Program for Algebraically Integrable Adjoint Foliated Structures.

Yifei Chen, Jihao Liu, and Yanze Wang. *International Mathematics Research Notices*, 2026(6): rnago45. 2026

(2) Flop between algebraically integrable foliations on potentially KLT varieties.

Yifei Chen, Jihao Liu, and Yanze Wang. *International Journal of Mathematics*, 36(11):2550035. 2025

(3) A Note on the Sarkisov Program.

Yifei Chen and Yanze Wang. *Higher Dimensional Algebraic Geometry: A Volume in Honor of V. V. Shokurov*, LMS Lecture Note Series, pp. 231–263, Cambridge University Press. 2025

AI4Math Practice

• **Rethlas** (github.com/hiraethecho/Rethlas): forked & extended a two-agent natural-language math-reasoning harness (generation + verification).

Projects

suckless X11 desktop components (dwm/dwmblocks) fork — enhanced by C patches, no AI.

lichtung Hugo theme (built via AI chat), powers two personal sites.

calman CLI task & event manager on iCalendar (vibe-coded, Rust).

markerss TUI RSS reader (vibe-coded, Rust/ratatui).

AiKit tools/skills/plugins for coding agents (esp. pi) + quick-deploy scripts.

Skills

- **Mathematics:** birational geometry — MMP, Sarkisov program, foliations.
- **Engineering:** Arch Linux (Niri/dwm) + Neovim, Git, C, LaTeX.
- **Languages:** Chinese (native), English (professional).